课堂笔记整理 5.19

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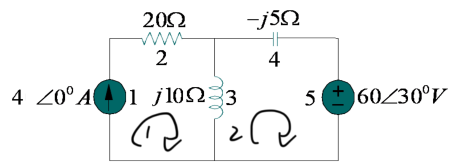
堂练1、

用网孔法，网孔1有 $I_1=4A$，网孔2有 $(j10-j5)I_2-j10I_1+60\angle30^\circ=0$

可解得 $I_2=10.58\angle79.1^\circ\ A$



① 对电流源：

$V_1=20I_1+j10(I_1-I_2)=80+j10(4-2-j10.39)=184.984\angle6.21^\circ V$

$P_1=-\frac{1}{2}\times184.984\times4\times\cos6.21^\circ=-367.8\ W$

② 对电阻：

$P_2=\frac{1}{2}80\times4=160\ W$

③ 对电容：

$I_2=10.58\angle79.1^\circ$

$V_4=-j5I_2=5\angle-90^\circ\times10.58\angle79.1^\circ=52.9\angle(79.1^\circ-90^\circ)$

$P_4=\frac{1}{2}52.9\times10.58\cos(-90^\circ)=0$

④ 对电感：

$I_3=I_1-I_2=2-j10.39=10.58\angle-79.1^\circ$

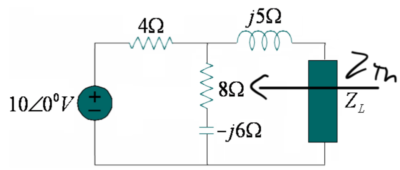
$V_3=j10(I_1-I_2)=105.8\angle(-79.1^\circ+90^\circ)$

$P_3=\frac{1}{2}\times105.8\times10.58\cos90^\circ=0$

⑤ 对电压源：

$P_5=\frac{1}{2}\times60\times10.58\cos(30^\circ-79.1^\circ)=207.8\ W$

堂练2



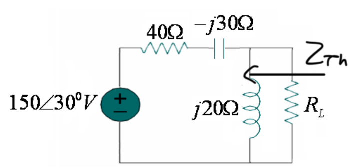
对负载两端求戴维南等效：

$V_{Th}=\frac{8-j6}{4+8-j6}\times10=7.454\angle-10.3^\circ\ V$

$Z_{Th}=j5+4\|(8-j6)=j5+\frac{4(8-j6)}{4+8-j6}=2.933+j4.467\ \Omega$

则 $P_{max}=\frac{|V_{Th}|^2}{8R_{Th}}=\frac{7.454^2}{8\times2.933}=2.368\ W$

堂练3



同样求 R_L 两端戴维南等效：

$Z_{Th}=j20\|(40-j30)=\frac{j20(40-j30)}{j20+40-j30}=9.412+j22.35\Omega$

$V_{Th}=\frac{j20}{j20+40-j30}(150\angle30^\circ)=72.76\angle134^\circ V$

$R_L=|Z_{Th}|=\sqrt{9.412^2+22.35^2}=24.25\ \Omega$

$I=\frac{V_{Th}}{Z_{Th}+R_L}=\frac{72.76\angle134^\circ}{33.39+j22.35}=1.8\angle100.2^\circ A$

$P_{max}=\frac{1}{2}|I|^2R_L=\frac{1}{2}\times1.8^2\times24.25=39.29\ W$

堂练4

(a) $V_{rms} = \frac{60}{\sqrt{2}}\angle -10^\circ$，$I_{rms} = \frac{1.5}{\sqrt{2}}\angle 50^\circ$

复功率：

$$\tilde{S} = V_{rms} I_{rms} = \frac{60}{\sqrt{2}}\angle -10^\circ \times \frac{1.5}{\sqrt{2}}\angle -50^\circ = 45\angle -60^\circ \ VA.$$

视在功率为：$S = |\tilde{S}| = 45\ VA$

(b) 由 $\tilde{S} = 45\angle -60^\circ = 45[\cos(-60^\circ) + j\sin(-60^\circ)] = 22.5 - j38.97$

由于 $\tilde{S} = P + jQ$，有功功率为 $P = 22.5\ W$.

无功功率为 $Q = -38.97\ VAR$

(c) 功率因数为：$pf = \cos(-60^\circ) = 0.5$

负载阻抗为：$Z = \frac{V}{I} = \frac{60\angle -10^\circ}{1.5\angle 50^\circ} = 10\angle -60^\circ\ \Omega$